Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

END SEMESTER PROJECT REVIEW

Date: 06-05-2026

TICKET BOOKING OF AN INTERNAL DEPARTMENT EVENT

SDG: SDG 8: Decent Work and Economic Growth

Presented By:

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Courses Handling Faculty:

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INTRODUCTION

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- Traditional event coordination often suffers from manual booking errors and data inconsistencies.
- EventHub provides a centralized digital platform to automate ticketing and resource management.
- This project is crucial for enhancing transparency and reducing administrative overhead for organizers
- .
- It features a modern Spring Boot backend synchronized with a high-performance React frontend.
- The system offers real-world relevance through its application in festivals, seminars, and workshops.
- A dynamic seat mapping engine ensures efficient venue utilization and prevents overbooking.
- Advanced Admin Dashboards provide real-time insights into event performance and ticket sales.
- Overall, it serves as a comprehensive solution for modern, secure, and streamlined event management.

PROBLEM STATEMENT

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- Currently, many event management processes are still handled using manual methods or siloed spreadsheets
- This leads to inconsistencies in attendee data and significant delays in ticket processing.
- Organizers lack real-time visibility into ticket sales and event metrics across the platform.
- Fragmented systems make it difficult for users to discover and book diverse events seamlessly.
- There is a high risk of ticket duplication and fraud without a centralized digital verification system.
- Communication between organizers and attendees regarding event updates is often slow and inefficient.
- Therefore, a reliable, automated, and centralized system is needed to improve overall performance and accuracy.

OBJECTIVE(S)

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- **To design and develop** a centralized ticketing platform called "Tech Fest 2026" that automates event discovery and booking.
- **To implement** a dynamic seat management system that ensures real-time availability and prevents overbooking.

PROPOSED SYSTEM

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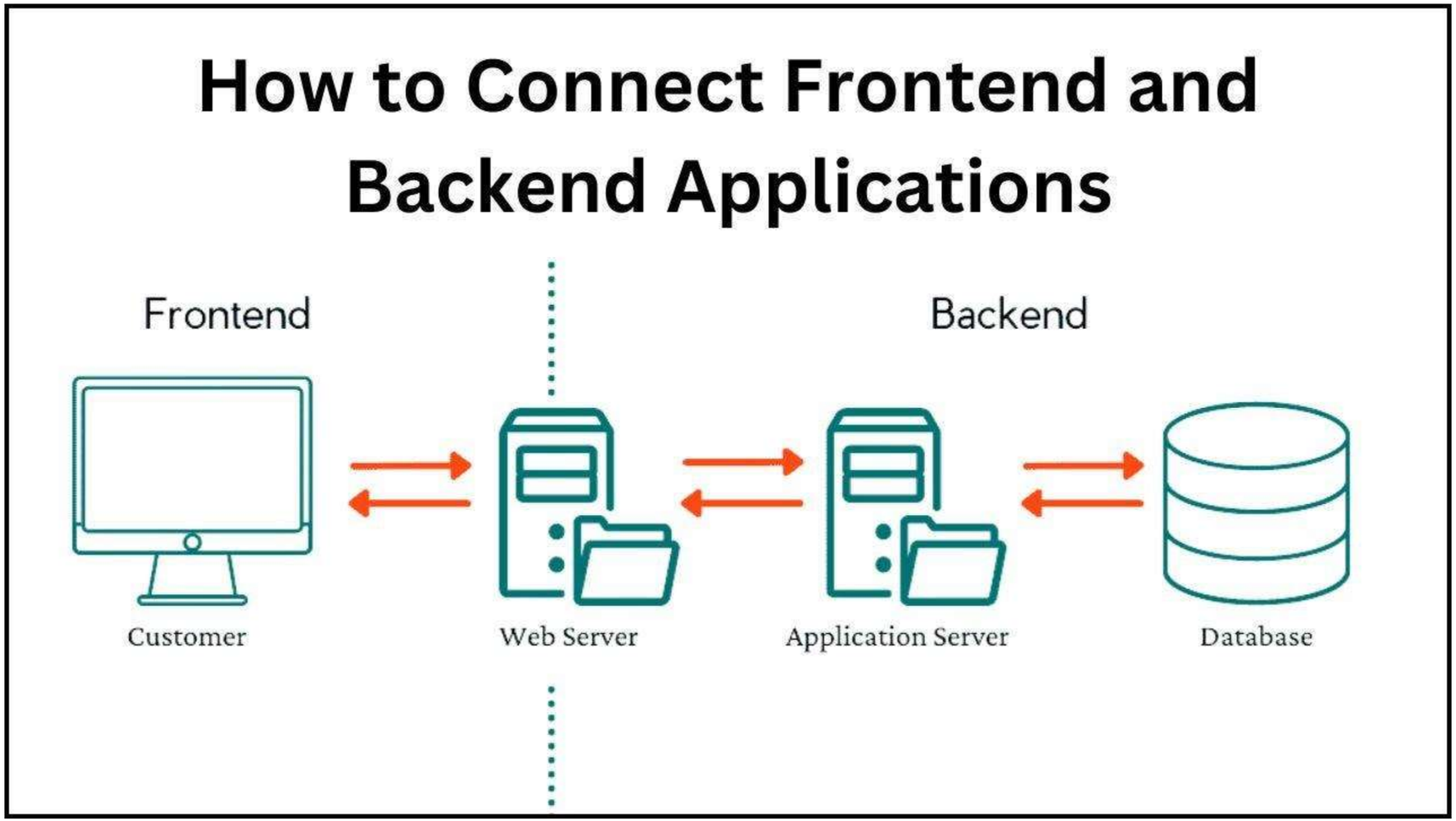
- Description: A full-stack web application leveraging Spring Boot for the backend and React for a modern, responsive frontend.
- Real-time Booking: Users can instantly browse available events and purchase tickets with live updates.
- Automated Seat Mapping: Implements a dynamic grid system (A, B... AA, AB) that handles varying venue capacities.
- Admin Dashboard: Provides powerful management tools for organizers to track sales and monitor attendee lists.
- Centralized Database: Uses H2/SQL to maintain a single source of truth for all events and transaction records.
- Secure Authentication: Features role-based access control to distinguish between attendees and administrators.
- Seamless UI/UX: Designed for a "premium" feel with smooth transitions, dark mode, and mobile-responsive layouts.
- Scalability: The modular architecture allows for the easy addition of new features and event categories.

PROPOSED SYSTEM ARCHITECTURE



WORKFLOW

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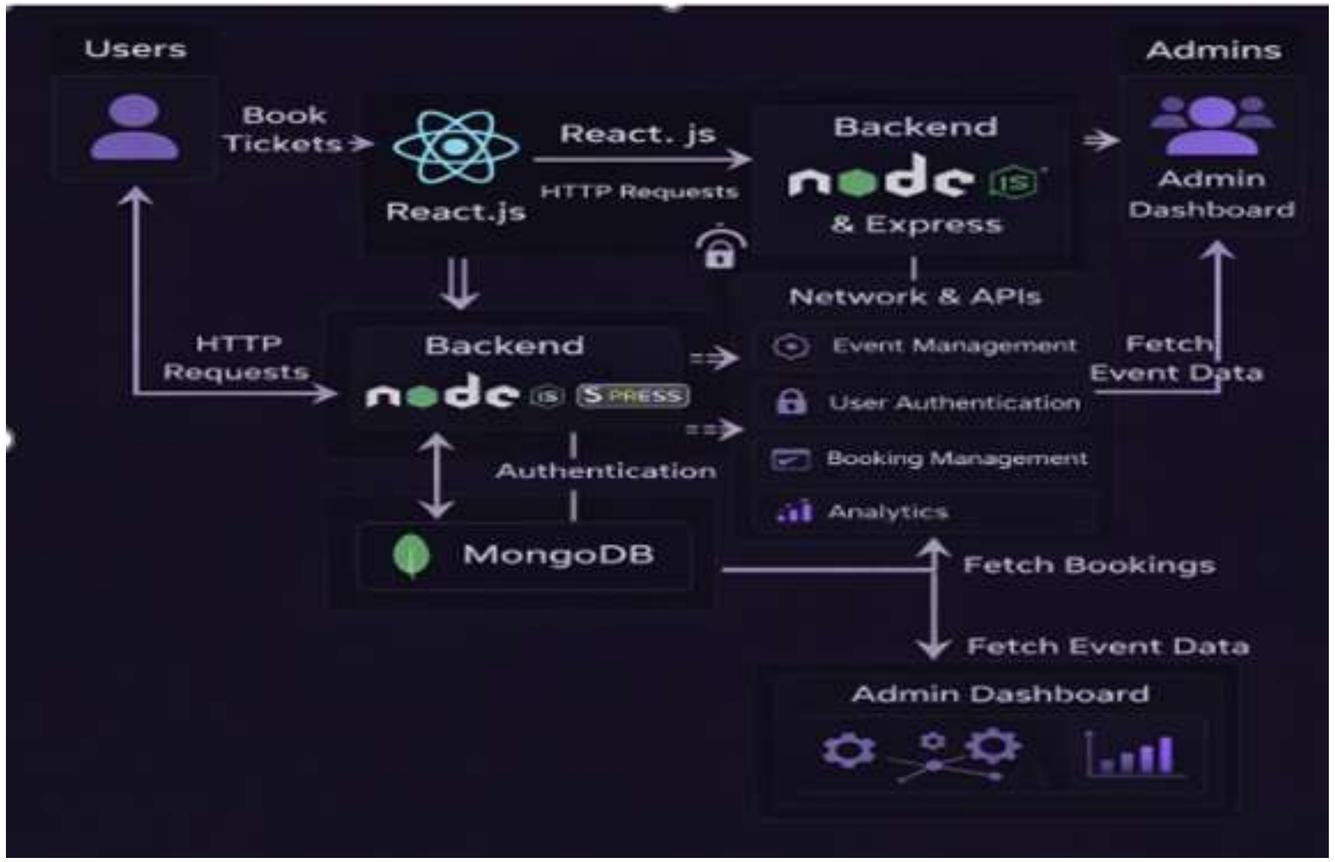

TECHNOLOGIES USED

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Frontend: React.js (v19), Tailwind CSS.

Backend: Java 17, Spring Boot 3.4.0, Maven, Spring Data JPA (Hibernate).

Database: H2 (High-performance, in-memory/file-based relational database).

Tools: VS Code (IDE), ECLIPSE.

Software Requirements: Windows 10/11, Node.js v20+, Java JDK 17, Modern Web Browser.

Hardware Requirements: 8GB RAM minimum, Intel Core i5 Processor or equivalent, 500MB Free Disk Space.

MODULE DESCRIPTION

Module 1: User Authentication & Profiles: Handles secure user registration and login, maintaining persistent sessions and distinguishing between regular users and administrative staff.

Module 2: Event Discovery & Grid Management: Implements the dynamic event showcase and the advanced seat selection engine featuring robust, automated row labeling.

Module 3: Transactional Booking Engine: Manages the core business logic for ticket reservations, ensuring real-time seat synchronization and preventing overbooking in the database.

Module 4: Admin Management Console: A dedicated module for platform administrators to create events, update ticket capacities, and monitor global booking statistics.

IMPLEMENTATION SCREENSHOTS

- A secure and modern admin login portal for Tech Fest 2026, designed with a sleek dark theme and authentication interface.
- It enables administrators to manage events, user bookings, and platform operations through a centralized control center.

IMPLEMENTATION SCREENSHOTS

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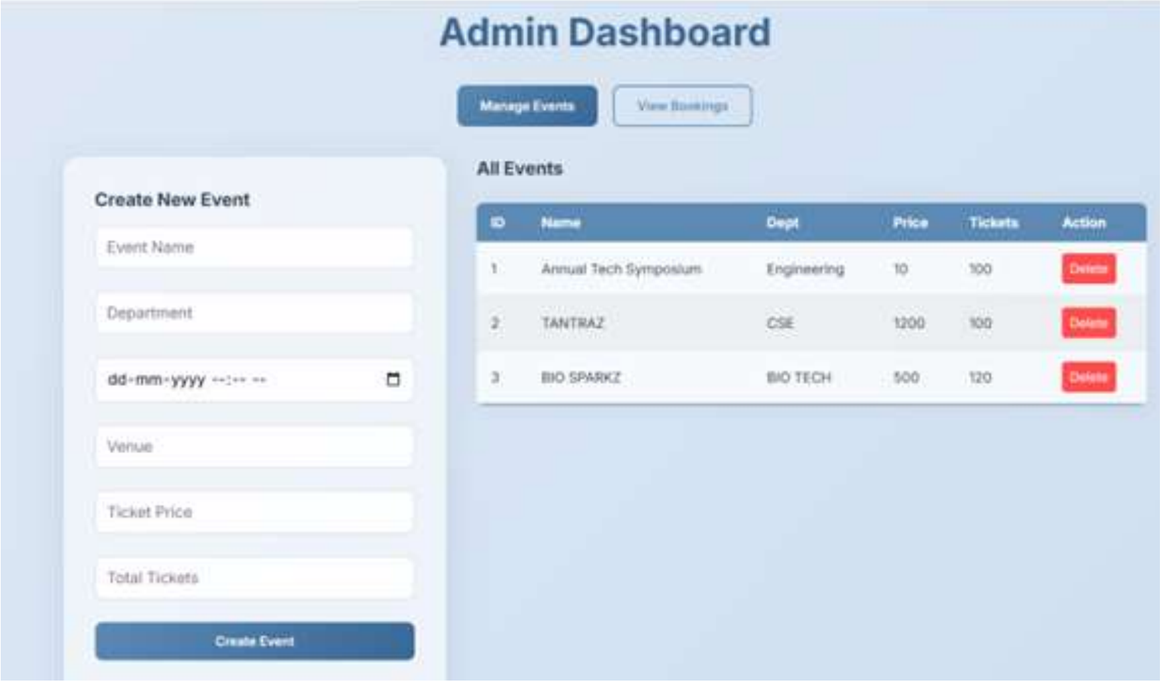

- A dynamic Tech Fest 2026 admin dashboard that allows organizers to add, manage, and monitor campus events efficiently.
- It features event creation forms and a real-time active events panel for seamless administration.

IMPLEMENTATION SCREENSHOTS

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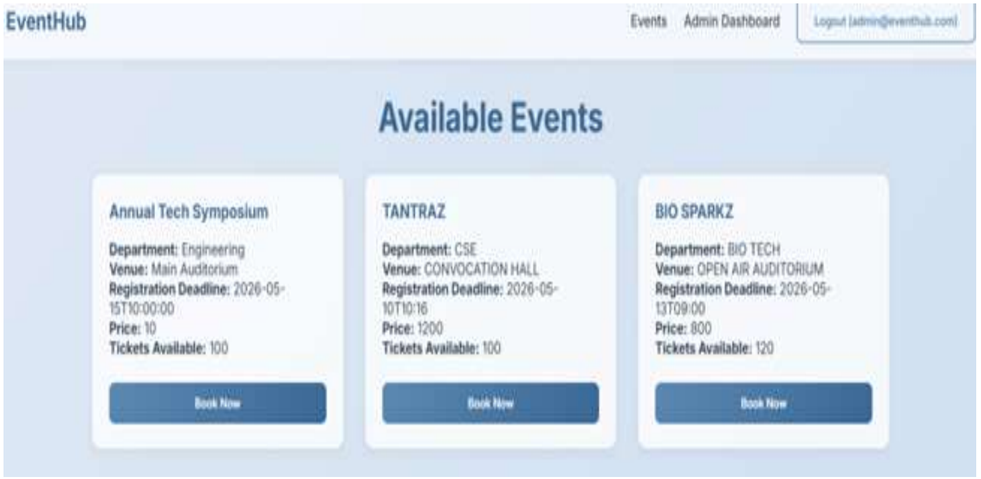
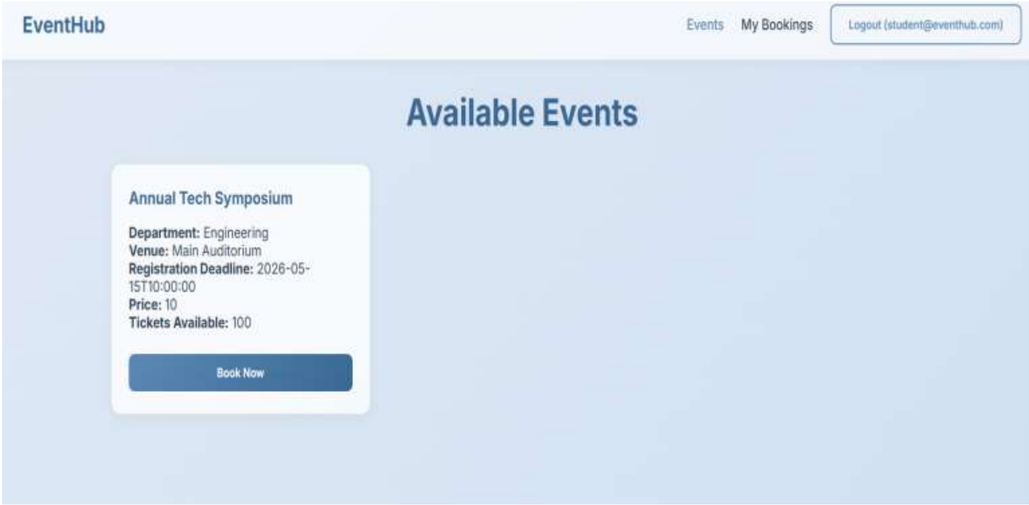
- Admin Login Page



- User Login Page



IMPLEMENTATION SCREENSHOTS

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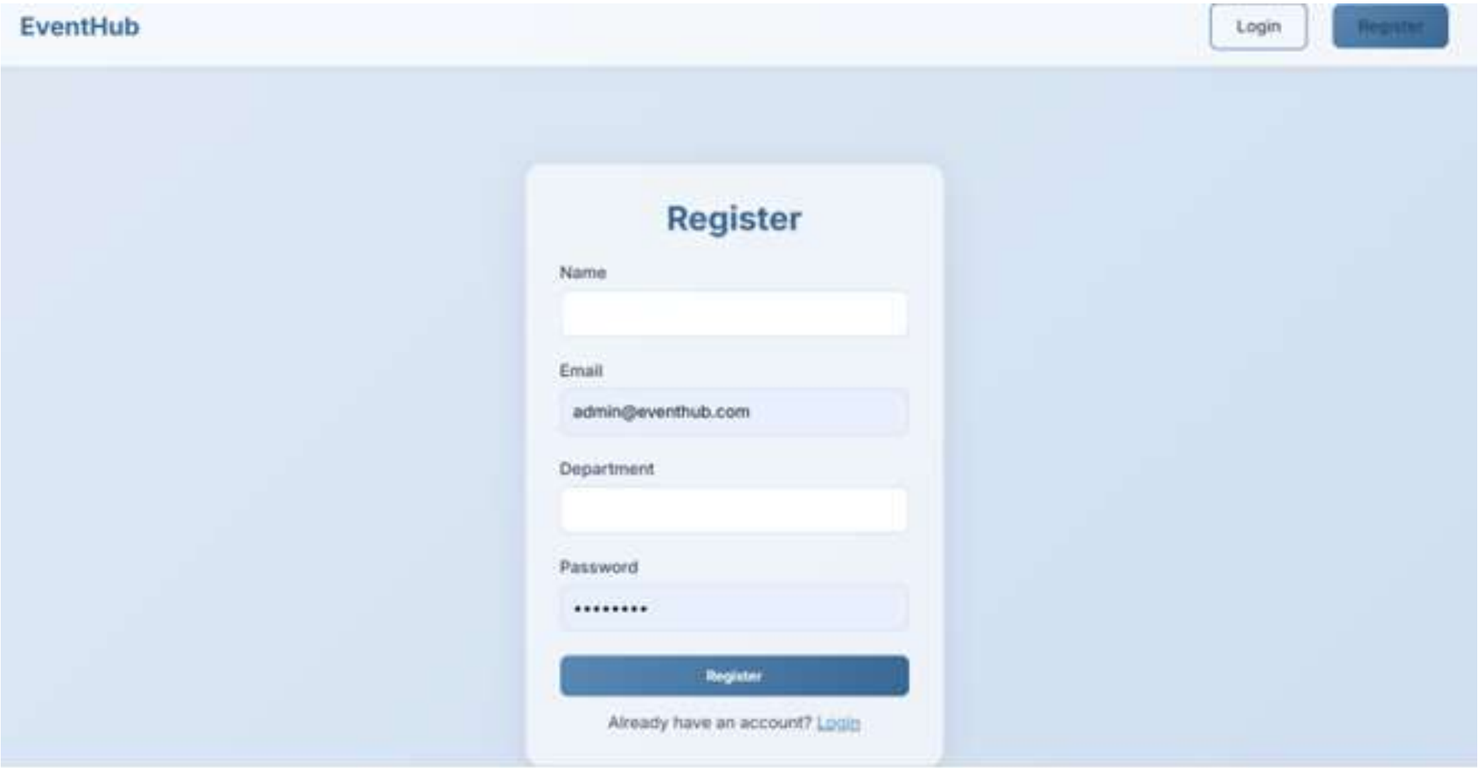
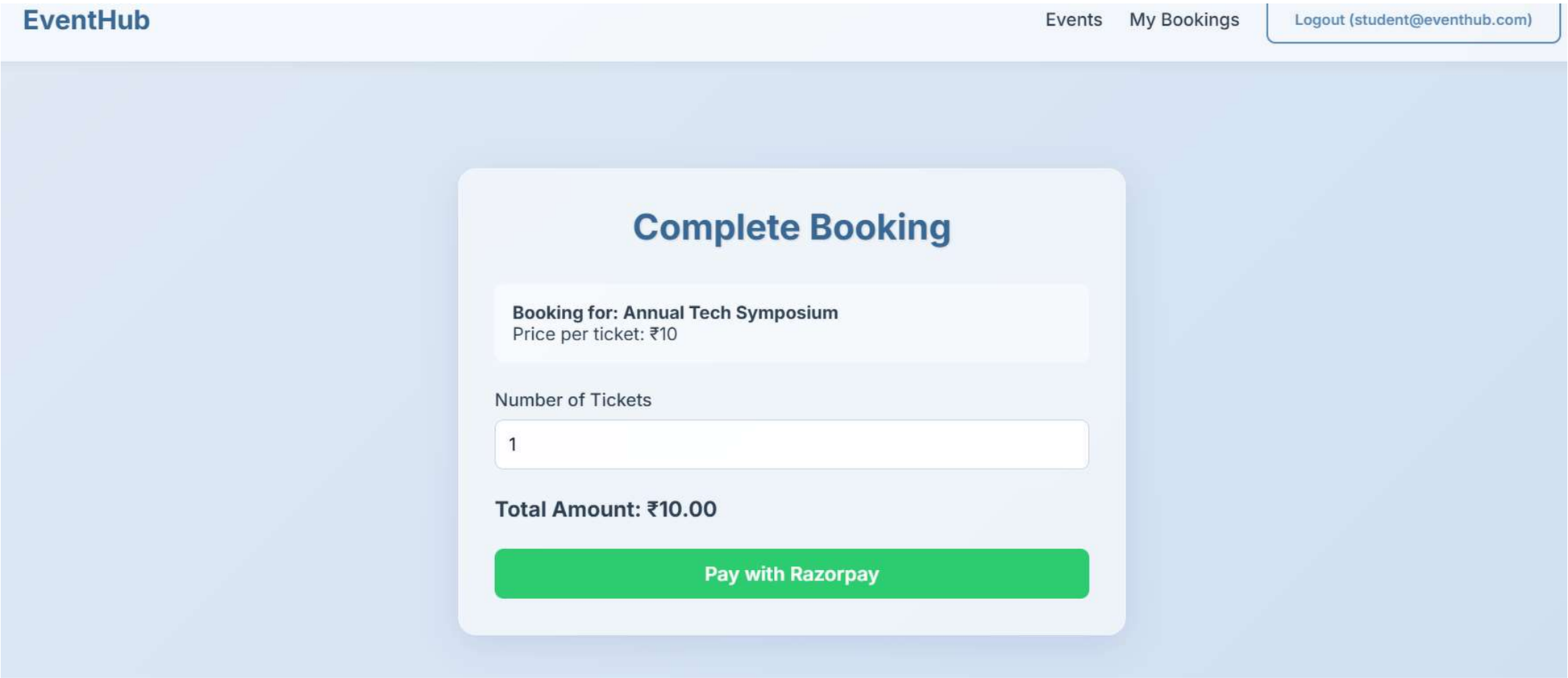


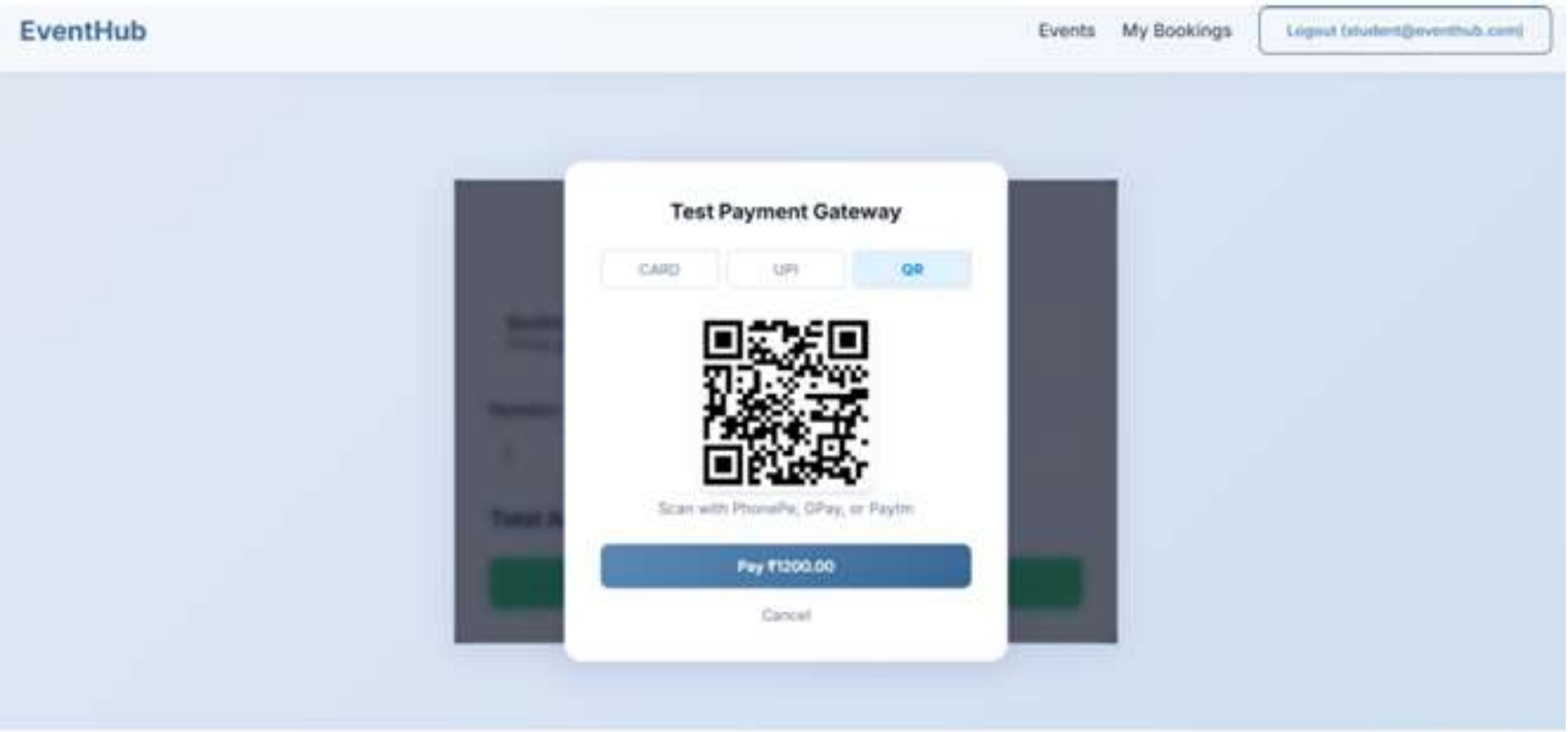
The image shows a user registration page for the EventHub platform with a clean and minimal design. It includes input fields for name, email, department, and password, along with a register button for account creation. Navigation options for login and registration are available at the top, ensuring easy access for users.

IMPLEMENTATION SCREENSHOTS



- The above interface represents the **payment module** of the Tech Fest 2026 event booking system. It provides a secure and structured environment for users to complete their ticket purchase for *Cultural Night 2026*. The design follows a modern dark-themed UI with a clear visual hierarchy to guide users through the payment process.

IMPLEMENTATION SCREENSHOTS



- The above interface represents the **UPI/QR payment module** of the Tech Fest 2026 ticket booking system. It provides a simple and efficient alternative payment method, allowing users to complete transactions using mobile-based UPI applications.

IMPLEMENTATION SCREENSHOTS

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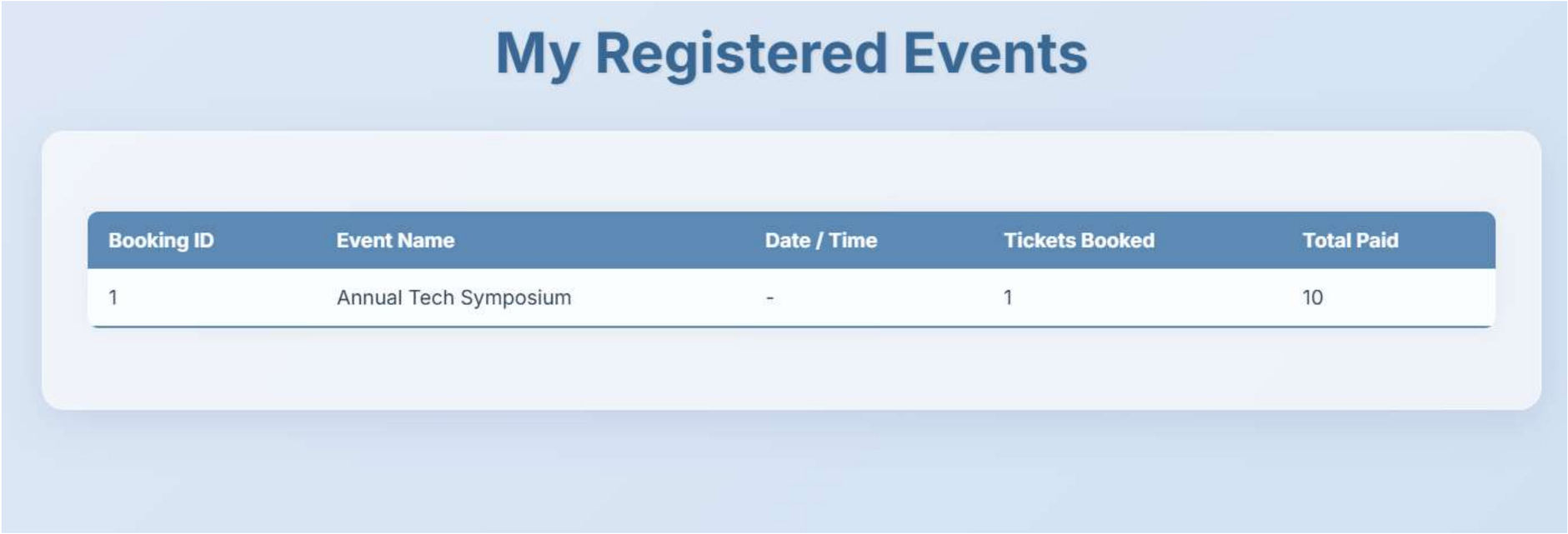

- The above interface represents the **booking confirmation module** of the Tech Fest 2026 ticket booking system. It is displayed after a successful payment, confirming that the user's tickets have been booked. The design maintains a consistent dark-themed UI with a centered confirmation card, ensuring clarity and focus.

ADVANTAGES & LIMITATIONS

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ADVANTAGES
•Complete Automation
•Real-time Synchronization
•Enhanced User Experience
•Centralized Data Management
•High Scalability
•Transparency and Trust

DISADVANTAGES (LIMITATIONS)
•Internet Dependency
•Infrastructure Requirements
•Local Storage Limitations
•Payment Gateway Complexity
•Technical Maintenance
•Learning Curve.

Thank you